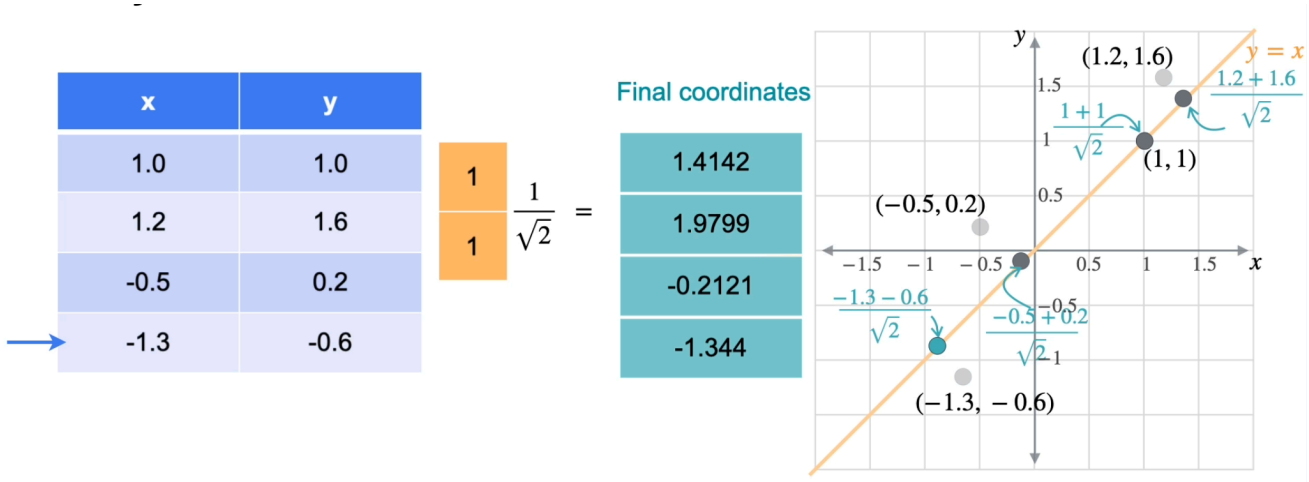
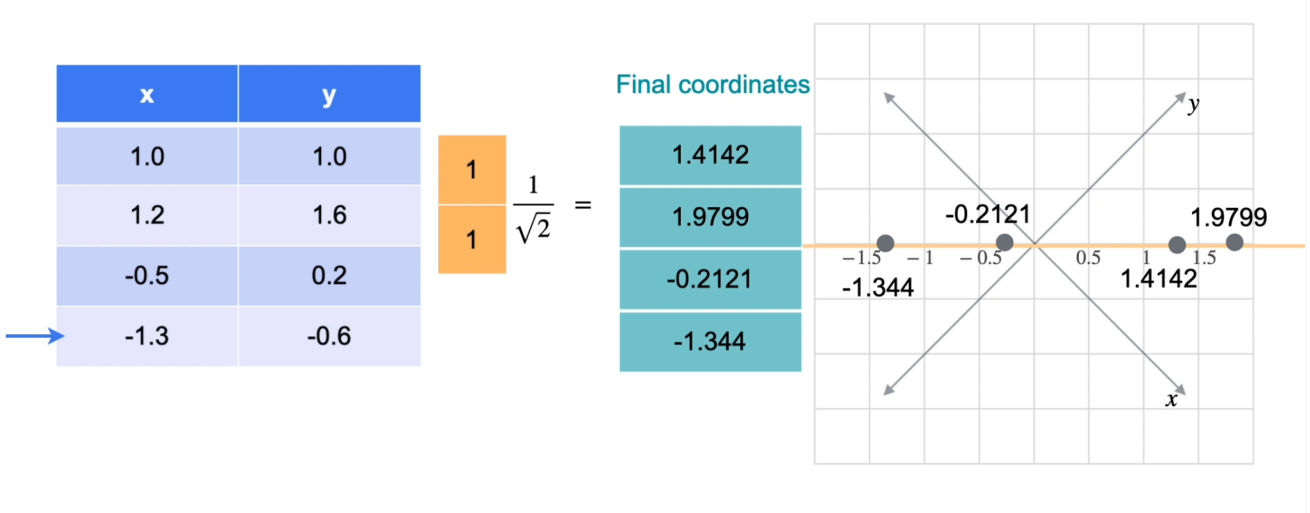
PCA

Benefits of PCA

- Reduce the dimension (columns) of the dataset
- Preserve as much information as possible
- Simpler Visualization

Projection



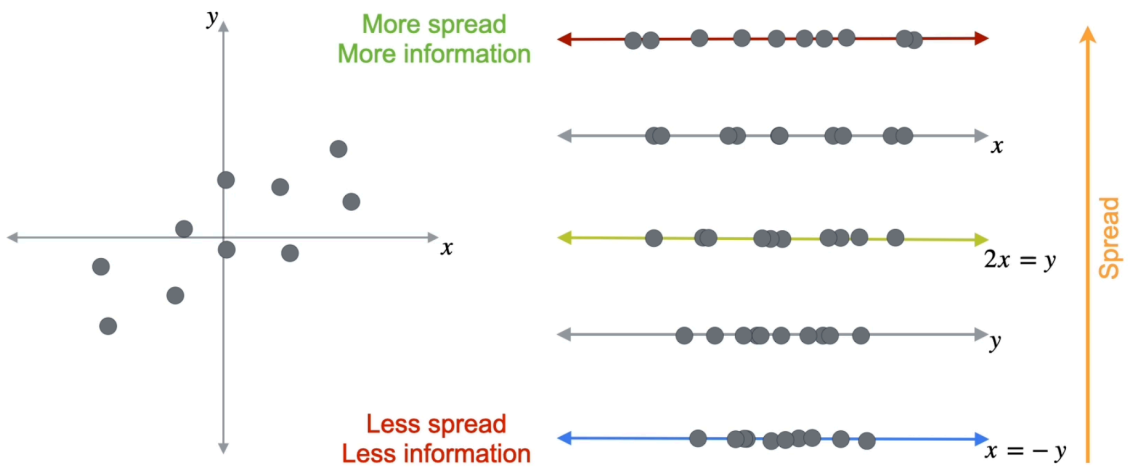


- To project a matrix A onto a vector v

$$A_p = A\frac{v}{||v||_2}$$

- It is important to keep track of Dimension: if A has $r \times c$ then the vector v has $c \times 1$ and finally the A_p has $r \times 1$
  - the same thing with larger dimension

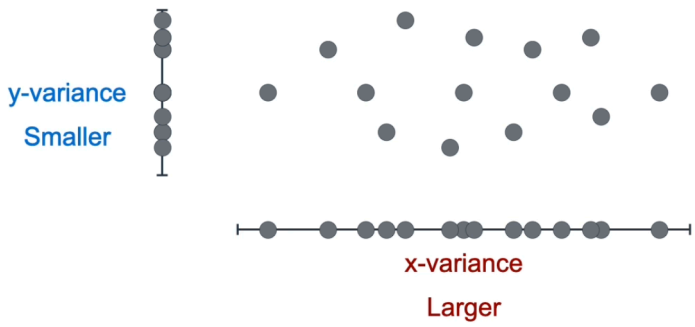
In PCA, preserving more spread, mean preserving more information



Goal: find the project that preserve most of the info

Variance

It describe how spread our data is.



$$Var(x) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

$$Var(y) = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2$$

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"The average square distance form the mean"
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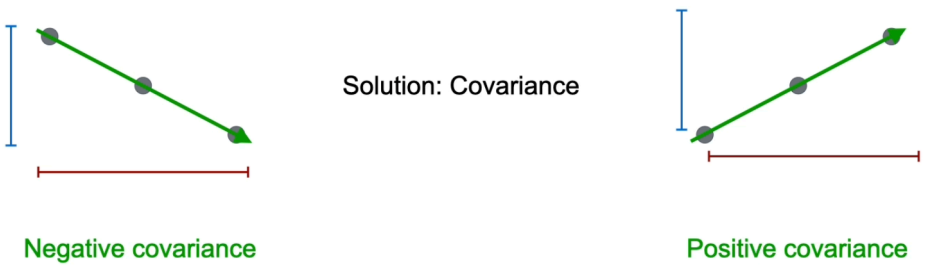
Problem with variance

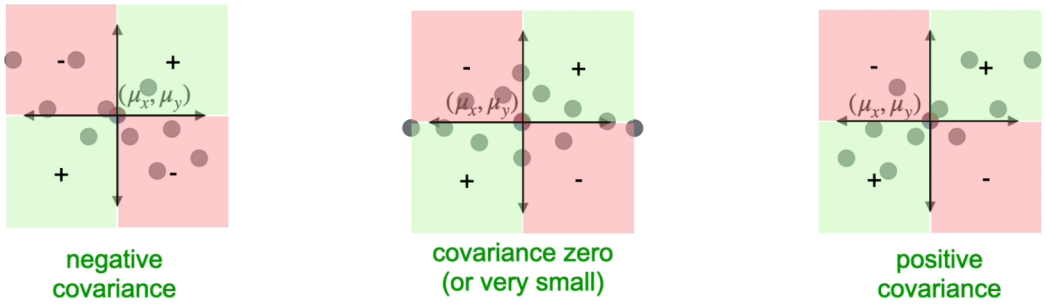


Covariance = how y changes according to x

Formula

$$Cov(x, y) = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \mu_x)(y - \mu_y)$$

"Take the average"

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"The direction of the relationship between two variables"
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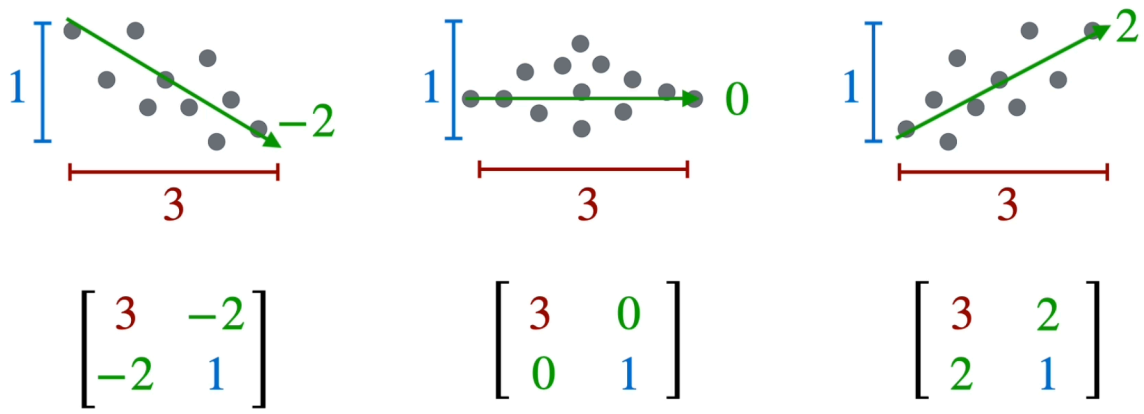
Covariance Matrix



Blue line = y variance
Red line = x variance

$$C = \frac{1}{n-1}(A - \mu)(A - \mu) = Var(x) \quad Cov(x, y) Cov(y, x) \quad Var(y)$$

$$Cov(x, x) = Var(x)$$

$$A = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_n & y_n \end{bmatrix} \qquad \mu = \begin{bmatrix} \mu_x & \mu_y \\ \mu_x & \mu_y \\ \vdots & \vdots \\ \mu_x & \mu_y \end{bmatrix}$$

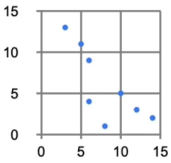
$$C = \frac{1}{n-1}(A-\mu)^T(A-\mu) = \frac{1}{n-1}\left(\begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_n & y_n \end{bmatrix} - \begin{bmatrix} \mu_x & \mu_y \\ \mu_x & \mu_y \\ \vdots & \vdots \\ \mu_x & \mu_y \end{bmatrix}\right)^T \left(\begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_n & y_n \end{bmatrix} - \begin{bmatrix} \mu_x & \mu_y \\ \mu_x & \mu_y \\ \vdots & \vdots \\ \mu_x & \mu_y \end{bmatrix}\right)$$

$$= \frac{1}{n-1}\begin{bmatrix} x_1-\mu_x & y_1-\mu_y \\ x_2-\mu_x & y_2-\mu_y \\ \vdots & \vdots \\ x_n-\mu_x & y_n-\mu_y \end{bmatrix}^T \begin{bmatrix} x_1-\mu_x & y_1-\mu_y \\ x_2-\mu_x & y_2-\mu_y \\ \vdots & \vdots \\ x_n-\mu_x & y_n-\mu_y \end{bmatrix}$$

$$C = \frac{1}{n-1}(A-\mu)^T(A-\mu) = \frac{1}{n-1}\left(\begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_n & y_n \end{bmatrix} - \begin{bmatrix} \mu_x & \mu_y \\ \mu_x & \mu_y \\ \vdots & \vdots \\ \mu_x & \mu_y \end{bmatrix}\right)^T \left(\begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \vdots \\ x_n & y_n \end{bmatrix} - \begin{bmatrix} \mu_x & \mu_y \\ \mu_x & \mu_y \\ \vdots & \vdots \\ \mu_x & \mu_y \end{bmatrix}\right)$$

$$= \frac{1}{n-1}\begin{bmatrix} x_1-\mu_x & y_1-\mu_y \\ x_2-\mu_x & y_2-\mu_y \\ \vdots & \vdots \\ x_n-\mu_x & y_n-\mu_y \end{bmatrix}^T \begin{bmatrix} x_1-\mu_x & y_1-\mu_y \\ x_2-\mu_x & y_2-\mu_y \\ \vdots & \vdots \\ x_n-\mu_x & y_n-\mu_y \end{bmatrix}$$

$$= \frac{1}{n-1}\begin{bmatrix} x_1-\mu_x & x_2-\mu_x & \dots & x_n-\mu_x \\ y_1-\mu_y & y_2-\mu_y & \dots & y_n-\mu_y \end{bmatrix} \begin{bmatrix} x_1-\mu_x & y_1-\mu_y \\ x_2-\mu_x & y_2-\mu_y \\ \vdots & \vdots \\ x_n-\mu_x & y_n-\mu_y \end{bmatrix} = \begin{bmatrix} Var(x) & Cov(x,y) \\ Cov(y,x) & \end{bmatrix}$$



Similar X and Y variances
Negative covariance

A

x	y
10	5
12	3
6	9
6	4
5	11
14	2
8	1
3	13

$\mu_x = 8 \quad \mu_y = 6$

$A - \mu$

$x-\mu_x$	$y-\mu_y$
2	-1
4	-3
-2	3
-2	-2
-3	5
6	-4
0	-5
-5	8

$\frac{1}{7}$

$(A-\mu)^T$

$x-\mu_x$	2	4	-2	-2	-3	6	0	-5
$y-\mu_y$	-1	-3	3	-2	5	-4	-5	8

$A - \mu$

$x-\mu_x$	$y-\mu_y$
2	-1
4	-3
-2	3
-2	-2
-3	5
6	-4
0	-5
-5	8

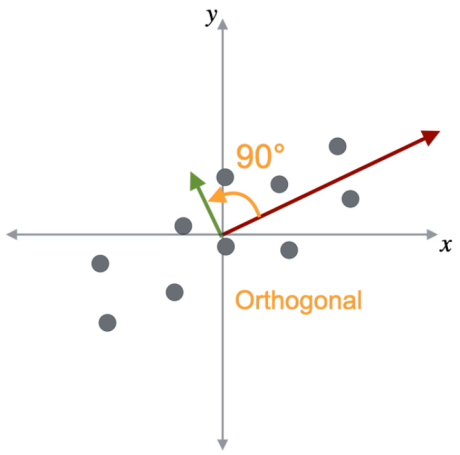
$$C = \begin{bmatrix} 14 & -11.86 \\ -11.86 & 19.71 \end{bmatrix}$$

PCA

Best line = Most variance

PCA combines:

- Projections
- Eigenvalues/Eigenvectors
- Covariance Matrix



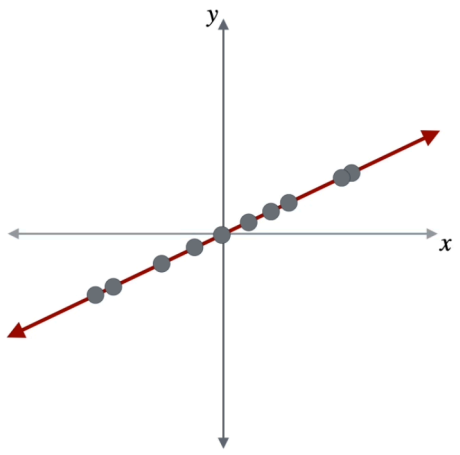
$$C = \begin{bmatrix} 9 & 4 \\ 4 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$ Eigenvectors (direction)

$$11 \quad 1$$ Eigenvalues (magnitude)

The eigenvector will always be orthogonal as long as C is symmetric (when transposed, the 4 changes spot)

We choose the one with the biggest eigenvalue (magnitude)



$$C = \begin{bmatrix} 9 & 4 \\ 4 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$ Eigenvectors (direction)

$$11$$ Eigenvalues (magnitude)

PCA Mathematical Formulation

You have n observations of 5 variables $(x_1, x_2, x_3, x_4, x_5)$ Goal: Reduce to 2 variables

1 Create matrix

5 variables

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{15} \\ x_{21} & x_{22} & \cdots & x_{25} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{n5} \end{bmatrix} \; n \text{ Observations}$$

2 Center the data

$$X - \mu = \begin{bmatrix} x_{11} - \mu_1 & x_{12} - \mu_2 & \cdots & x_{15} - \mu_5 \\ x_{21} - \mu_1 & x_{22} - \mu_2 & \cdots & x_{25} - \mu_5 \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} - \mu_1 & x_{n2} - \mu_2 & \cdots & x_{n5} - \mu_5 \end{bmatrix}$$

3 Calculate Covariance Matrix

$$C = \frac{1}{n-1}(X-\mu)^T(X-\mu) = \begin{bmatrix} Var(X_1) & Cov(X_1,X_2) & Cov(X_1,X_3) & Cov(X_1,X_4) & Cov(X_1,X_5) \\ Cov(X_1,X_2) & Var(X_2) & Cov(X_2,X_3) & Cov(X_2,X_4) & Cov(X_2,X_5) \\ Cov(X_1,X_3) & Cov(X_2,X_3) & Var(X_3) & Cov(X_3,X_4) & Cov(X_3,X_5) \\ Cov(X_1,X_4) & Cov(X_2,X_4) & Cov(X_3,X_4) & Var(X_4) & Cov(X_4,X_5) \\ Cov(X_1,X_5) & Cov(X_2,X_5) & Cov(X_3,X_5) & Cov(X_4,X_5) & Var(X_5) \end{bmatrix}$$

4 Calculate Eigenvectors and Eigenvalues

Big	λ_1	v_1
	λ_2	v_2
	λ_3	v_3
	λ_4	v_4
Small	λ_5	v_5

5 Create Projection Matrix

$$V = \begin{bmatrix} \frac{v_1}{\|v_1\|_2} & \frac{v_2}{\|v_2\|_2} \end{bmatrix}$$

6 Project Centered Data

$$X_{PCA} = (X-\mu)V$$